

BYUNGJIN JUN

ASSISTANT PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE, KOREA MILITARY ACADEMY

Seoul, Republic of Korea | E-mail: bjun@kma.ac.kr | Homepage: <https://byungjinjun.github.io>

EDUCATION

Northwestern University, Evanston, IL

- » *Ph.D., Computer Science* 2023
- » *M.S., Computer Science* 2014
- » Advisor: Fabián E. Bustamante

Korea Military Academy, Seoul, Republic of Korea

- » *Bachelor of Science, Information Science and Military Science* 2009

PROFESSIONAL EXPERIENCE

Chair, *Department of Computer Science, Korea Military Academy (KMA)* Jul. 2026 ~ Present

Assistant Professor, *Department of Computer Science, KMA* Aug. 2023 ~ Present

- » Teaching and conducting research in computer networks, security, and AI

Research Intern, *Telefónica Research, Barcelona* Oct. ~ Dec. 2018, Nov. 2019 ~ Feb. 2020

- » Researched Roaming of Things and IPX

Full-time Instructor, *Department of Computer Science, KMA* Aug. 2014 ~ Jul. 2017

Signal Officer, *Republic of Korea Army (ROKA)* Mar. 2009 ~ Aug. 2012

- » Network Operations officer: Operated large-scale communication networks of an Army Division
- » Signal Platoon Leader: Led a 30-soldier platoon providing deployable wireless communications

RESEARCH INTERESTS

Computer Networks and Networked Systems; Network Security; Web Performance and QoE; Defense and Mission-Critical Systems

CURRENT PROJECTS

Mobile Security Architecture Design for Defense 5G Infrastructure, ROKA HQ Jul. 2026 ~ Present

- » Principal Investigator

AI Geo-Agent-Based Platform for Tourism Safety and Border Security Management in the Civilian Control Zone, Ministry of Culture, Sports, and Tourism Apr. 2026 ~ Present

- » Researcher

Robust Mobile Edge-Cloud Intelligence for Autonomous Manned-Unmanned Teaming in Disrupted Network Environments, Institute of Information & communications Technology Planning & Evaluation (IITP) Apr. 2026 ~ Present

- » Researcher

Dynamic Trust Connectivity and Intelligent Control for Multi-Layer, Multi-Tier Networks, IITP Apr. 2025 ~ Present

- » Researcher

Improving Users' Web Experience on Any Network of Device, National Research Foundation of Korea (NRF) Apr. 2024 ~ Present

- » Principal Investigator

AI-based situation awareness and decision support for counter-terrorism, Agency for Defense Development Dec. 2023 ~ Present

- » Researcher

COMPLETED PROJECTS

Information Security Classification and Protection Measures for Data-Driven Defense Systems, ROKA HQ

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» Researcher Mar. 2026 ~ Aug. 2026

Improving Access Control in the Civilian Control Zone, Ministry of National Defense

» Researcher Jul. 2025 ~ Dec. 2025

Decentralized Identifier based access control system for Army, Seoul Business Agency

» Lead Researcher, Demonstration Site Sep. 2024 ~ Aug. 2025

QUIC for the challenged networks, Researcher

» Built a proxy server that translates TCP traffic on ground to QUIC for in-flight/satellite users Jan. 2019 ~ Aug. 2019

Requirement analysis of integrating military weapon system networks for the future Tactical Communications Network, Ministry of National Defense

» Analyzed network systems connecting sensors and shooters
» The umbrella project of [Publications No.5](#) Dec. 2016 ~ Oct. 2017

Utilization of Google Glass for advertisements on the street, Developer

» Developed server-client system (Django/Android for Google Glass) Feb. 2014 ~ Aug. 2014

PUBLICATIONS

19. Rethinking Web QoE : A Threshold-Based View of User Abandonment

May. 2026

» *Seunghyun Lee, Byungjin Jun, Hyoyoung Lim*
» In Proceedings of the 2026 KSII Spring Conference

18. Proposal of Threshold-Based Survival Metric for Mission-Critical QoE Evaluation in FPV Drones

» *Junyeol Park, Byungjin Jun, Hyoyoung Lim* May. 2026
» In Proceedings of the 2026 KSII Spring Conference

17. Initial Usage Patterns of a DID-Based Access Control System in a Military Facility

May. 2026

» *Byungjin Jun*
» In Proceedings of the 2026 KSII Spring Conference

16. Measuring Webpage Visual Correctness as Users Experience It: Dynamic Region-aware Visual Evaluation

Apr. 2026

» *Gitaek Lee, Taehwa Lee, Byungjin Jun*
» In Proceedings of ACM CHI Extended-Abstract 2026

15. Structural Security Analysis of Military DID-based Access Control Systems using Zero Trust Principles

Jan. 2026

» *Chaemin Jeon, Byungjin Jun*
» In Proceedings of 2026 KICS Winter Conference

14. Network-Separated Security and Traffic-Governance Architecture for Internet-Based Military DID Access Control Systems

Jan. 2026

» *Byungjin Jun, Kyuyong Shin, Hyeondon Choi*
» The Journal of Korean Institute of Communications and Information Sciences (J-KICS)

13. Scenario Design Methodology for Applying AI Prediction in Security Operations Without Failure Data

» *Byungjin Jun, et. al.* Jun. 2025
» In Proceedings of Korea Institute of Science and Technology 2025

12. (Book) Cyber Warfare 101, 1st Edition (Korean)

Dec. 2024

» *J. Yoo, et al, including Byungjin Jun*

11. Reining in Mobile Web Performance with Document and Permission Policies

Mar. 2022

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- » **Byungjin Jun**, Fabián E. Bustamante, Ben Greenstein, Ian Clelland
- » In Proceedings of ACM HotMobile 2022

10. Networked systems as witnesses: association between content demand, human mobility and an infection spread Oct. 2021

- » Sana Asif, **Byungjin Jun**, Fabián E. Bustamante, John P. Rula
- » In Proceedings of ACM IMC 2021

9. WebTune: A Distributed Platform for Web Performance Measurements Sep. 2021

- » **Byungjin Jun**, Matteo Varvello, Yasir Zaki, and Fabián E. Bustamante
- » In Proceedings of IFIP TMA 2021

8. A first look at the IP eXchange Ecosystem Oct. 2020

- » Andra Lutu, **Byungjin Jun**, Fabián E. Bustamante, Diego Perino, Marcelo Bagnulo, and Carlos Gamboa Bontje
- » Sigcomm Computer Communication Review, Volume 50 Issue 4 (October 2020)

7. Where Things Roam: Uncovering Cellular IoT/M2M Connectivity Oct. 2020

- » Andra Lutu, **Byungjin Jun**, Alessandro Finamore, Fabián E. Bustamante, and Diego Perino
- » In Proceedings of ACM IMC 2020

6. AMP Up Your Mobile Web Experience: Characterizing the Impact of Google's Accelerated Mobile Project Oct. 2019

- » **Byungjin Jun**, Fabián E. Bustamante, S. Y. Whang, and Z. S. Bischof
- » In Proceedings of ACM MobiCom 2019

5. An Analysis of IPv6 Transition Status and Efficient Address Allocation for the Defense Information Systems Oct. 2017

- » **Byungjin Jun**, K. Shin, J. Kang, J. Yoo
- » Korean Journal of Military Arts and Science. Volume 73 Issue 2, pp. 227~248

4. (Book) Introduction to Java Programming, 2nd Edition (Korean) Jan. 2017

- » J. Yoo, et al, including **Byungjin Jun**
- » Chapter 1. Basic concept of computer programming

3. A Study on the Current Status and Future Direction of Cyber Warfare Education in KMA toward Effective Conduct of Cyber Warfare Jun. 2016

- » Kyuyong Shin, **Byungjin Jun**, Jungcho Kang, Pokki Park, Insoo Lee, Jincheol Yoo
- » Korean Journal of Military Arts and Science. Volume 72 Issue 2, pp. 131~167

2. (Book) Introduction to Cyber Warfare, 2nd Edition (Korean) Feb. 2016

- » M. Oh, et al, including **Byungjin Jun**
- » Chapter 6. Comparative study of capabilities in cyber warfare, Chapter 9. Technologies of cyber threats

1. Mobile AD(D): Estimating Mobile App Session Times for Better Ads Feb. 2015

- » John P. Rula, **Byungjin Jun**, Fabián E. Bustamante
- » In Proceedings of HotMobile 2015

SELECTED TEACHING EXPERIENCE

Artificial Intelligence (Capstone), KMA Spring 2026

- » Designed and taught

Operating Systems, KMA Fall 2024, Fall 2025

- » Designed and taught

Intro to Cyber Warfare, KMA Fall 2014, Fall 2015, Fall 2016, Spring 2025 ~ Spring 2026

- » Co-Instructor

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Intro to Computer Programming, KMA » Co-Instructor	Spring 2024, Fall 2026
Discrete Mathematics, KMA » Designed and taught	Fall 2023, Spring 2024, Fall 2024
Distributed Systems, Northwestern University » Teaching Assistant	Spring 2019, Spring 2020, Spring 2021
Intro to Computer Science, KMA » Co-Instructor	Fall 2014, Spring 2015, Fall 2015, Fall 2016, Spring 2017
Data Structures, KMA » Designed and taught	Spring 2015, Spring 2017

SELECTED AWARDS AND HONORS

2021 Google PhD Fellowship in Systems and Networking, Google	2021
First place Award in Signal Officers' Advanced Course, <i>Korean Army Chief of Staff</i>	2016
Commendation for the extensive contribution to promote KJMAS to a KCI journal, <i>President of KMA</i>	2015
Commendation for establishing stable network in the annual Joint Forces Exercise, <i>20th Division Commander</i>	2011
Summa Cum Laude, KMA	2009